Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/05/2023
PI Name:	Nusbaum, Derek M.D., Ph.D.		
Project Title:	Comparison of the Negative Effects of Carbon Dioxide Exposure Among Ground, Hypobaric, and Spaceflight Environments		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Principal Investigator (PI) is also affiliated with the University of Texas Medical Branch.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/04/2022	**End Date:**	01/14/2024
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 01/14/2024 as Dr. Garbino took over the project in January 2024 (Ed., 4/18/24)		
Key Personnel Changes/Previous PI:	NOTE: Dr. Nusbaum has left the laboratory conducting this research. See project with same title under Dr. Stijn Thoolen for later reporting (Ed., 1/14/24) September 2023 update per the PI (Ed., 9/22/23): 1) Michael Bishop, Ph.D. (KBR/NASA Johnson Space Center) is no longer a Co-Investigator with the project. 2) Mathias Basner, M.D., Ph.D. (University of Pennsylvania) has been added to the project as a Co-Investigator.		
COI Name (Institution):	Abercromby, Andrew Ph.D. (NASA Johnson Space Center) Scully, Robert Ph.D. (NASA Johnson Space Center) Levine, Benjamin M.D. (University of Texas Southwestern Medical Center) Babb, Tony Ph.D. (University of Texas Southwestern Medical Center) Pawelczyk, James Ph.D. (Pennsylvania State University) Baughman, Alexander M.S. (NASA Johnson Space Center) Basner, Mathias M.D., Ph.D. (University of Pennsylvania)		
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Task Description:

Carbon dioxide (CO2) produced through the metabolism of astronauts inside space suits will accumulate to levels that affect health and performance if the space suit does not provide an adequate CO2 removal capability. Suit helmet geometry, vent configurations, vent gas flow rates, and metabolic rate all affect the extent to which CO2 is successfully removed from the vicinity of the user's nose and mouth and, ultimately, the level of CO2 (PiCO2) that the user inspires. During certain suited contingencies, such as failure of the CO2 removal system, a purge valve must be opened in certain planetary EVA suits, which in turn affects the efficiency with which CO2 is removed from the helmet. Computational fluid dynamics modeling of these contingency scenarios suggests that PiCO2 levels exposed to the crew could be as high as 17mmHg.
NASA-Standard-3001 Volume 2 Rev B includes a standard for inspired CO2 during nominal suited operations, which was based on characterization of PiCO2 in the existing Extravehicular Mobility Unit (EMU) spacesuit. However, no suited PiCO2 standard exists for suited contingency operations due to a lack of directly applicable data. The nominal standard had been adopted as the nominal PiCO2 requirement for the exploration EMU space suit (known as xEMU). In the absence of a contingency CO2 standard, the xEMU project and follow-on projects have been using the upper bound of the nominal PiCO2 requirement (15mmHg) as the contingency requirement. Because of this, as currently designed, there are potential failure modes for certain planetary EVA suits that would violate this 15mmHg contingency requirement. Meeting this 15mmHg requirement may ultimately require a redesign of developing hardware, potentially leading to cost and schedule overruns.

While some limited functional performance data does exist for PiCO2 exposures greater than 15mmHg, there is uncertainty and inconsistency within the ground-based data on the performance impacts of these elevated CO2 levels, especially when in association with exercise. Some studies suggest that CO2 levels of 20mmHg or higher are tolerable without major performance impacts, while other studies suggest that these CO2 levels could be unacceptably impacting. Thus, the human health and performance risks associated with CO2 exposure in the range of 15 to 20mmHg or higher in the flight environment remain an area of debate.

The CO2 Walk Back Study (the companion study to this study) will characterize CO2 related symptoms and functional performance on a simulated EVA contingency walk back task across a range of PICO2 levels. A known limitation of the CO2 Walk Back Study is that subjects will not be in the hypobaric spaceflight environment that exists in a spacesuit. There are potential physiologic differences between normobaria and hypobaria independent of gas partial pressures. As an example, there is evidence of differing physiologic responses to normobaric hypoxia as compared to hypobaric hypoxia. Similarly, the concern has been raised that the hypobaric environment may compound the effects of elevated CO2 exposure (in ways not fully understood, and independent of PICO2) when compared to the normobaric environment. The purpose of this study is to quantify the extent to which responses to CO2 are affected by reduced pressure and the spaceflight environment. The high-fidelity EVA contingency walk back simulation task is not feasible to implement in the hypobaric and spaceflight environments, so this study will use alternative, validated cognitive performance tasks while recording the same symptomatic data as the CO2 Walk Back Study. Results will be used to either validate that CO2 dose-response relationships from the CO2 Walk Back Study are directly applicable to contingency EVA scenarios or else provide a recommended quantitative adjustment to dose-response relationships to account for the effects of reduced pressure and/or spaceflight.

This study will use a repeated measures ground, hypobaric chamber, and flight study design. The control arm of this study will be performed in a shirtsleeve environment in a laboratory setting. The flight arm of this study will be performed in a shirtsleeve environment on the ISS. The chamber arm of this study will be performed in a shirtsleeve environment in a hypobaric chamber, operating at 222.4mmHg (4.3 psia). The hypobaric exposure arm of this study will take place in the 20' Chamber in Building 7 of Johnson Space Center. The spaceflight exposure arm of this study will take place with US Crew Members during their missions aboard the International Space Station. For all arms of this study, subjects will breath compressed air (Flight Arm) or 100% oxygen (Chamber Arm) mixed with the prescribed fraction of CO2. Subjects will be asked to cycle on a cycle ergometer (CEVIS in-flight) at a self-selected cadence within a defined range, bounded by the metabolic equivalent of a 3km/hr walking speed at a minimum and the metabolic rate of 1600btu/hr at a maximum. Cognitive performance will be quantified using the validated, Cognition battery of cognitive performance tests. Physical performance will be measured by evaluating the duration cycled, up to a maximum of one hour. The severity of specific, CO2 related symptoms (headache, dizziness, nausea, numbness, tingling, shortness of breath, air hunger, fatigue, euphoria) will be quantified via survey prior to the testing session, at 10-minute intervals during the testing session, and immediately following the testing session, as well as every 10 minutes for one hour post exposure to document recovery. Throughout the course of the trial, metabolic rate, respiratory rate, heart rate, tidal volume, minute ventilation, expiratory CO2, and transcutaneous CO2 will be measured from the test subject continuously.

This study will be conducted using a total of 15 different, non-astronaut (but matched on key demographics) subjects for the Chamber arm and 5 mission-assigned Astronauts for the Flight Arm. Each subject will be exposed to 7 different partial pressures of carbon dioxide (0, 5, 10, 15, 20, 25, 30mmHg) for 1 hour at a time, each on a different day to provide sufficient time for washout of exposure. During the one hour of exposure, the subjects will breathe compressed air (Flight) or 100% oxygen (Chamber) +CO2 via oronasal facemask. The order in which subjects are exposed to each of the different carbon dioxide partial pressures will utilize balanced randomization and the subjects will be blinded to the PiCO2 levels. The Chamber Arm of this study will be an 8-day, 24 hour a day, live-in chamber study. The subjects in the Chamber Arm will live at an atmosphere of 8.2psia and 34% oxygen for the duration of their stay in the chamber. Each day following a pre-breathe, the chamber environment will decrease to 4.3psia and 100% oxygen for the subjects to perform their simulated "walk back". Each day, the subjects will be exposed to a different partial pressure of carbon dioxide during their "walk back".

This study will determine the maximum inspired CO2 level that still enables crewmembers to complete a 1-hour contingency walk back within acceptable functional performance limits. Specifically, results will inform future planetary EVA suit design and operations and will be incorporated as a Contingency Inspired Carbon Dioxide Standard for Suited Operations in NASA-STD-3001. Furthermore, as this study will include the physiologic, symptomatic, and performance effects of CO2 at nominal partial pressures of PiCO2 in addition to contingency levels, this study will also provide data to inform and clarify overall risk of CO2 exposure in the flight environment, not just for suited, EVA operations, which has been deemed an important human health and performance risk for future spaceflight missions.

Rationale for HRP Directed Research:	This project is directed because it contains highly constrained research. This project is a companion to its counterpart “Effect of Carbon Dioxide Exposure on Physical and Cognitive Performance in a Simulated Spaceflight Contingency Scenario”. This flight arm is necessary to extrapolate the results of the Rapid Response ground study to the altered gravity and atmospheric environments expected during exploration missions and should therefore be directed to the same PI team.
Research Impact/Earth Benefits:	NASA receives improved information on the effects of elevated CO2 exposure during spaceflight which will help determine tolerable limits that can be used for updating standards for both suited operations and habitat operations and inform future hardware design. This will enable safe and efficient exploration EVA activities and habitat design. NASA receives an understanding about the physiological responses to elevated carbon dioxide during contingency operations. Society benefits from the disseminated knowledge, including and especially how elevated CO2 may impact exercise capability, which may be important in various disease conditions such as COPD and asthma.
Task Progress:	The Institutional Review Board (IRB) protocol for the Flight Arm of this study was developed over the course of the summer of 2022 and was approved by the NASA IRB on 7/26/22 (eIRB#: STUDY00000502). Subsequently, ROI has been working over the course of FY23 on their Feasibility Assessment for flight. The results of this assessment are expected to be released in Q4 of this fiscal year. The IRB protocol was initially put on hold for the Chamber Arm of this study while lessons learned from initial Exploration Atmosphere chamber runs, specifically regarding the ability to maintain chamber oxygen concentrations to within +/- 1%, were incorporated. The IRB protocol for the Chamber Arm was developed over the summer of 2023 and has now been submitted. It is currently awaiting evaluation by the board and approval. Work has been ongoing over Q2 and Q3 of this fiscal year with EC to develop chamber requirements for this study. Current work is being done to organize the first chamber data run for this study, with dry runs, equipment testing, and Test Conductor training to occur in Q1 of FY24 during the next Exploration Atmosphere chamber run in October/November 2023. This will set up a subsequent chamber run for this study in Spring 2024. NOTE: Continued by "Comparison of the Negative Effects of Carbon Dioxide Exposure Among Ground, Hypobaric, and Spaceflight Environments" (PI: Thoolen) due to Dr. Nusbaum's departure from the laboratory conducting this research. See Dr. Thoolen's project for additional reporting.
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